

Trigonometry

ANSWER KEY



Right-triangle trigonometric ratios

- 1 In a right triangle, angle θ has an opposite side of 5 and hypotenuse 13. Find $\sin \theta$ and the length of the adjacent side.
$$\text{Adjacent} = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12. \sin \theta = \frac{5}{13}.$$
 - 2 In a right triangle, angle θ has an opposite side of 15 and hypotenuse 17. Find $\cos \theta$.
$$\text{Adjacent} = \sqrt{17^2 - 15^2} = \sqrt{289 - 225} = \sqrt{64} = 8. \cos \theta = \frac{8}{17}.$$
 - 3 In a right triangle, angle θ has an adjacent side of 7 and hypotenuse 25. Find $\sin \theta$ and $\tan \theta$.
$$\text{Opposite} = \sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24. \sin \theta = \frac{24}{25}, \tan \theta = \frac{24}{7}.$$
 - 4 In a right triangle, the two legs are 20 and 21, and the hypotenuse is 29. Find $\sin \theta$ and $\cos \theta$ for the angle θ opposite the side of length 20.
$$\sin \theta = \frac{20}{29}, \cos \theta = \frac{21}{29}.$$
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Special angles

- 5 State the exact value of $\sin(30^\circ)$.
$$\sin(30^\circ) = \frac{1}{2}.$$
 - 6 State the exact value of $\cos(45^\circ)$.
$$\cos(45^\circ) = \frac{\sqrt{2}}{2}.$$
 - 7 State the exact value of $\tan(60^\circ)$.
$$\tan(60^\circ) = \sqrt{3}.$$
 - 8 State the exact values of $\sin(90^\circ)$ and $\cos(0^\circ)$.
$$\sin(90^\circ) = 1 \text{ and } \cos(0^\circ) = 1.$$
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Converting between radians and degrees

- 9 Convert 180° to radians.
$$180^\circ = \pi \text{ radians.}$$
- 10 Convert 90° to radians.
$$90^\circ = \frac{\pi}{2} \text{ radians.}$$
- 11 Convert $\frac{5\pi}{6}$ radians to degrees.
$$\frac{5\pi}{6} \times \frac{180^\circ}{\pi} = 150^\circ.$$

- 12 Convert 270° to radians.

$$270^\circ = \frac{3\pi}{2} \text{ radians.}$$

The Pythagorean identity

- 13 Given $\sin \theta = \frac{3}{5}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.
 $\cos^2 \theta = 1 - \frac{9}{25} = \frac{16}{25}$. Since θ is acute, $\cos \theta = \frac{4}{5}$.

- 14 Given $\cos \theta = \frac{5}{13}$ with θ acute, find $\sin \theta$.
 $\sin^2 \theta = 1 - \frac{25}{169} = \frac{144}{169}$. Since θ is acute, $\sin \theta = \frac{12}{13}$.

- 15 Given $\sin \theta = 0.6$, find $\cos^2 \theta$.
 $\cos^2 \theta = 1 - 0.36 = 0.64$.
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Simplifying trigonometric expressions

- 16 Simplify $\sin^2 \theta + \cos^2 \theta + 1$.
 $\sin^2 \theta + \cos^2 \theta = 1$, so the expression equals $1 + 1 = 2$.

- 17 Simplify $\frac{1 - \cos^2 \theta}{\sin \theta}$ (for $\sin \theta \neq 0$).
 $1 - \cos^2 \theta = \sin^2 \theta$, so the expression equals $\frac{\sin^2 \theta}{\sin \theta} = \sin \theta$.

- 18 Simplify $\tan \theta \cos \theta$ (for $\cos \theta \neq 0$).
 $\tan \theta = \frac{\sin \theta}{\cos \theta}$, so $\tan \theta \cos \theta = \sin \theta$.
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Graphs: amplitude and period

- 19 State the amplitude and period of $y = 5\sin(x)$.
Amplitude: 5. Period: 2π .

- 20 State the amplitude and period of $y = 3\cos(2x)$.
Amplitude: 3. Period: $\frac{2\pi}{2} = \pi$.

- 21 State the amplitude and period of $y = 2\sin\left(\frac{x}{3}\right)$.
Amplitude: 2. Period: $\frac{2\pi}{1/3} = 6\pi$.

- 22 A sine graph has amplitude 4 and period π . Write its equation in the form $y = A\sin(Bx)$.
 $A = 4$. $B = \frac{2\pi}{\pi} = 2$. So $y = 4\sin(2x)$.
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Graphs: phase shift and vertical shift

- 23 The graph of $y = \sin(x)$ is shifted $\frac{\pi}{3}$ to the right and 2 units up. Write the new equation.
 $y = \sin\left(x - \frac{\pi}{3}\right) + 2$.
- 24 State the vertical shift and midline of $y = 3\cos(x) + 5$.
Vertical shift: up 5. Midline: $y = 5$.
- 25 The graph of $y = \cos(x)$ is shifted $\frac{\pi}{4}$ to the left. Write the new equation.
 $y = \cos\left(x + \frac{\pi}{4}\right)$.
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Solving basic trigonometric equations

- 26 Solve $2\sin\theta = 1$ for $0^\circ \leq \theta < 360^\circ$.
 $\sin\theta = \frac{1}{2}$. Solutions: $\theta = 30^\circ$ or $\theta = 150^\circ$.
- 27 Solve $\sqrt{2}\cos\theta = 1$ for $0^\circ \leq \theta < 360^\circ$.
 $\cos\theta = \frac{\sqrt{2}}{2}$. Solutions: $\theta = 45^\circ$ or $\theta = 315^\circ$.
- 28 Solve $\tan\theta = -1$ for $0^\circ \leq \theta < 360^\circ$.
Solutions: $\theta = 135^\circ$ or $\theta = 315^\circ$.
- 29 Solve $2\cos\theta + 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.
 $\cos\theta = -\frac{1}{2}$. Solutions: $\theta = 120^\circ$ or $\theta = 240^\circ$.
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Reference angles in other quadrants

- 30 Find $\sin(150^\circ)$ using the reference angle.
The reference angle is 30° ; in Quadrant II, sine is positive. $\sin(150^\circ) = \sin(30^\circ) = \frac{1}{2}$.
- 31 Find $\cos(240^\circ)$ using the reference angle.
The reference angle is 60° ; in Quadrant III, cosine is negative. $\cos(240^\circ) = -\cos(60^\circ) = -\frac{1}{2}$.
- 32 Find $\tan(300^\circ)$ using the reference angle.
The reference angle is 60° ; in Quadrant IV, tangent is negative. $\tan(300^\circ) = -\tan(60^\circ) = -\sqrt{3}$.
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Complementary angle relationships

- 33 Given $\sin(20^\circ) \approx 0.342$, use the relationship $\sin\theta = \cos(90^\circ - \theta)$ to find $\cos(70^\circ)$.
 $\cos(70^\circ) = \sin(20^\circ) \approx 0.342$.

- 34 Given $\sin(35^\circ) \approx 0.574$, find $\cos(55^\circ)$.
 $\cos(55^\circ) = \sin(35^\circ) \approx 0.574$.
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The unit circle: coordinates of key points

- 35 State the coordinates $(\cos \theta, \sin \theta)$ of the point on the unit circle at $\theta = 60^\circ$.
 $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$.
- 36 State the coordinates of the point on the unit circle at $\theta = 180^\circ$.
 $(-1, 0)$.
- 37 State the coordinates of the point on the unit circle at $\theta = 270^\circ$.
 $(0, -1)$.
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Word problems: angles of elevation

- 38 This question has two parts. A drone hovers directly above a landmark. From a point 200 meters from the base of the landmark, the angle of elevation to the drone is 35° .
- a Write an equation relating the drone's height h to the given information, using the tangent ratio.
 $\tan(35^\circ) = \frac{h}{200}$, so $h = 200\tan(35^\circ)$.
- b Find h , to the nearest meter, given $\tan(35^\circ) \approx 0.700$. $h \approx 200(0.700) = 140$ meters.
- 39 This question has two parts. A ladder 10 meters long leans against a wall, making an angle of 65° with the ground.
- a Write an equation for the height h the ladder reaches up the wall, using the sine ratio. $\sin(65^\circ) = \frac{h}{10}$, so $h = 10\sin(65^\circ)$.
- b Find h , to the nearest tenth of a meter, given $\sin(65^\circ) \approx 0.906$. $h \approx 10(0.906) = 9.1$ meters.